

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 30 — Hydrocarbons (Arenes)

Paper 4 Topical Examination Pack & Comprehensive Examiner Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701) — Paper 4
TOPIC COVERAGE	Benzene Structure & Delocalisation · Electrophilic Aromatic Substitution Mechanisms · Methylbenzene Reactions
COMPONENT	Paper 4 — A Level Structured Questions (Theory & Calculations)
TOTAL MARKS	220 Marks (50 Comprehensive Questions)

HOW TO USE THIS A LEVEL PAPER 4 PACK

1. Complete each structured question thoroughly in the answer spaces provided, adhering strictly to mark allocations.
2. Questions are classified into Core and Advanced tiers, covering multi-step chemical derivations, mechanisms, and numerical proofs.
3. Every question features authentic Cambridge A Level Paper 4 examination references (e.g. 9701/42/M/J/23) for benchmark cross-referencing.
4. Full point-by-point examiner mark schemes with step-by-step working and error-carried-forward (ECF) criteria are detailed at the end.

SYLLABUS SUBTOPICS COVERED

- 30.1 Benzene Structure, Bonding & Resonance Stability** — Kekulé structure versus delocalised π -system; planar regular hexagonal geometry, sp^2 hybridisation
- 30.2 Electrophilic Aromatic Substitution Reactions & Mechanisms** — Electrophilic substitution of electrophiles; nitration (conc. HNO_3 + conc. H_2SO_4 at 50–55 °C to generate NO_2^+);
- 30.3 Methylbenzene: Activating & Directing Effects** — Electron-donating inductive (+I) effect of alkyl groups; enhanced electron density in the aromatic ring
- 30.4 Side-Chain Oxidation of Alkylarenes** — Oxidation of alkyl side-chains (methylbenzene, ethylbenzene) to benzoic acid by prolonged reflux
- High-Frequency Core Repeats (Q41–Q50)** — The 10 most frequently tested Cambridge Paper 4 questions on Arenes from the past 10 years.

SECTION A: 6-MARK EXTENDED EXAM QUESTIONS (40% TARIFF · Q1–Q16)

1 Benzene Bonding, Enthalpy of Hydrogenation & Resonance Stability — 9701/42/M/J/23/Q6 [6 Marks]

HARD · Syllabus 30.1

The structure of benzene is explained by a delocalised π -electron system rather than the theoretical Kekulé structure (cyclohexa-1,3,5-triene).

The thermochemical comparison of their enthalpies of hydrogenation is depicted in Fig. 1.1.

Data: H°_{hyd} (cyclohexene) = -120 kJ mol^{-1} ; H°_{hyd} (benzene) = -208 kJ mol^{-1} .

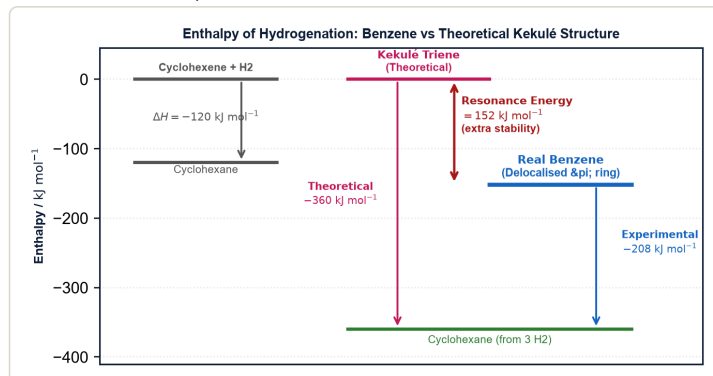


Fig. 1.1: Enthalpy of hydrogenation energy levels illustrating the 152 kJ mol^{-1} resonance stabilization energy of benzene.

- (a) Calculate the theoretical enthalpy of hydrogenation for cyclohexa-1,3,5-triene and deduce the resonance (delocalisation) energy of benzene. [2]

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- (b) Describe the bonding in benzene in terms of orbital hybridisation, σ -bonds, and delocalised π -orbitals. [2]

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- (c) State two physical or chemical pieces of experimental evidence (other than enthalpy of hydrogenation) that disprove the Kekulé structure. [2]

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2 Electrophilic Aromatic Substitution: Nitration of Benzene Mechanism — 9701/41/M/J/23/Q6 [6 Marks]

HARD · Syllabus 30.2

Benzene is nitrated by heating with a mixture of concentrated nitric acid and concentrated sulfuric acid at $50-55^\circ\text{C}$.

The general mechanism of electrophilic substitution is shown in Fig. 2.1.

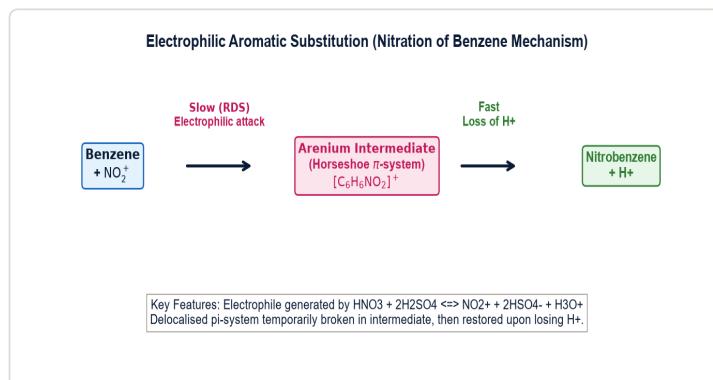


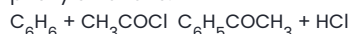
Fig. 2.1: Electrophilic aromatic substitution mechanism showing generation of NO_2^+ and the Wheland carbocation intermediate.

- (a) Write an equation for the generation of the nitronium ion electrophile, NO_2^+ , and state the catalytic role of concentrated sulfuric acid. [2]
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- (b) Draw the full mechanism for the nitration of benzene, showing curly arrows, the horseshoe arenium intermediate with positive charge, and the regeneration of the catalyst. [3]
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- (c) Explain why the temperature must be kept below 55 °C during this reaction. [1]
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3 Friedel-Crafts Acylation & Alkylation of Benzene — 9701/42/O/N/23/Q6 [6 Marks]

HARD · Syllabus 30.2

Benzene reacts with ethanoyl chloride, CH_3COCl , in the presence of anhydrous aluminium chloride, AlCl_3 , to produce phenylethanone:



- (a) Write an equation showing how the acylium electrophile, CH_3CO^+ , is generated by reaction with AlCl_3 . [2]
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- (b) Draw the mechanism for this acylation reaction, using curly arrows and showing the intermediate. [2]
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- (c) Explain why Friedel-Crafts acylation stops cleanly at mono-substitution, whereas Friedel-Crafts alkylation frequently leads to poly-alkylation. [2]
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4 Methylbenzene: Ring Substitution vs Side-Chain Free-Radical Chlorination — 9701/41/O/N/23/Q6 [6 Marks]

HARD · Syllabus 30.3

Methylbenzene, $\text{C}_6\text{H}_5\text{CH}_3$, reacts with chlorine under two completely different sets of conditions:

- Reaction 1: Cl_2 in the presence of anhydrous AlCl_3 in the cold, dark.
- Reaction 2: Cl_2 gas bubbled into boiling methylbenzene in the presence of UV light.

- (a) Identify the organic products formed in Reaction 1, naming the type of reaction mechanism and explaining the positions of substitution. [3]
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- (b) Identify the organic product formed in Reaction 2, naming the type of reaction mechanism. [2]
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- (c) Write equations for the propagation steps in the mechanism of Reaction 2. [1]
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5 Side-Chain Alkyl Oxidation to Benzoic Acid — 9701/42/M/J/22/Q6 [6 Marks]

HARD · Syllabus 30.4

Alkylarenes undergo side-chain oxidation when heated under prolonged reflux with alkaline potassium manganate(VII), KMnO_4 , followed by acidification with dilute sulfuric acid.

- (a) State the observations during the reflux with alkaline KMnO_4 and upon subsequent addition of dilute acid to the cooled mixture. [2]
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- (b) Propylbenzene, $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_3$, is heated with alkaline KMnO_4 and then acidified. Identify the organic product and any inorganic carbon by-product. [2]
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- (c) Explain why (1,1-dimethylethyl)benzene (tert-butylbenzene), $\text{C}_6\text{H}_5\text{C}(\text{CH}_3)_3$, does NOT undergo oxidation under these conditions. [2]
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6 Bromination of Benzene vs Phenol vs Methylbenzene — 9701/41/M/J/22/Q6 [6 Marks]

HARD · Syllabus 30.2

The reactivity of aromatic rings towards electrophilic bromination varies substantially:

- Compound A: Benzene
- Compound B: Methylbenzene
- Compound C: Phenol

- (a) Rank these three compounds in order of increasing reactivity towards electrophilic bromination (least reactive first). [1]
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- (b) State the reagents and conditions required to brominate benzene, writing the overall equation. [2]
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- (c) Explain why phenol reacts instantly with aqueous bromine without needing a catalyst, whereas benzene requires liquid bromine and an FeBr_3 catalyst. [3]
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7 Directing Effects: Nitration of Nitrobenzene vs Methylbenzene — 9701/42/O/N/22/Q6 [6 Marks]

HARD · Syllabus 30.3

Substituents on a benzene ring influence both the rate of further substitution and the position at which incoming electrophiles attack.

- (a) Explain why the nitro group ($-\text{NO}_2$) is deactivating and directs incoming electrophiles to the 3-position (meta-directing). [3]

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- (b) Draw the structural formula of the organic product obtained when nitrobenzene is heated with concentrated HNO_3 and concentrated H_2SO_4 at 100°C . [1]

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- (c) Contrast the reaction conditions required to nitrate nitrobenzene with those required to nitrate benzene. [2]

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8 Halogenation of Arenes: FeBr_3 vs UV Light Mechanisms — 9701/41/O/N/22/Q6 [6 Marks]

HARD · Syllabus 30.2

Ethylbenzene, $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_3$, reacts with bromine under different conditions to yield two structural isomers.

- (a) When ethylbenzene reacts with Br_2 and FeBr_3 in the dark, two isomeric bromo compounds are formed. Identify both products and state their positions of substitution. [2]

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- (b) When ethylbenzene is boiled with Br_2 in the presence of UV light, (1-bromoethyl)benzene is formed as the major product. Draw its structure and identify the chiral center. [2]

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- (c) Explain why bromination in UV light occurs preferentially at the C1 (benzylic) position rather than C2. [2]

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9 Combustion & Enthalpy of Arenes: Benzene vs Hexane — 9701/42/M/J/21/Q6 [6 Marks]

HARD · Syllabus 30.1

When benzene burns in air, it produces a very smoky, sooty yellow flame, whereas hexane burns with a much cleaner flame.

- (a) Write the balanced chemical equation for the complete combustion of liquid benzene. [2]

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- (b) Calculate the percentage by mass of carbon in benzene (C_6H_6) and in hexane (C_6H_{14}). [2]
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- (c) Explain why aromatic hydrocarbons burn with a characteristic smoky, luminous flame with unburnt carbon soot. [2]
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10 Synthesis of Benzoic Acid from Benzene via Multi-Step Route — 9701/41/M/J/21/Q6 [6 Marks]

HARD · Syllabus 30.4

A chemical manufacturer synthesizes benzoic acid, C_6H_5COOH , starting from benzene, C_6H_6 .
Route: Benzene Methylbenzene Benzoic acid.

- (a) State the reagent, catalyst, and reaction conditions required for Step 1 (Benzene Methylbenzene). [2]
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- (b) State the reagents and conditions required for Step 2 (Methylbenzene Benzoic acid). [2]
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- (c) Explain why an alternative route via bromobenzene and a Grignard reagent or nitrile is less efficient. [2]
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11 Aromatic vs Aliphatic Halogen Compounds: Chlorobenzene vs Chloroethane — 9701/42/O/N/21/Q6 [6 Marks]

HARD · Syllabus 30.2

Chloroethane hydrolyses readily when refluxed with aqueous NaOH, but chlorobenzene is completely inert under the same conditions.

- (a) Write the balanced chemical equation for the hydrolysis of chloroethane. [1]
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- (b) Explain why the carbon-chlorine bond in chlorobenzene is significantly stronger and shorter than in chloroethane. [3]
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- (c) State a second reason why nucleophilic attack by OH^- on chlorobenzene is prevented. [2]
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12 Friedel-Crafts Alkylation & Carbocation Rearrangement — 9701/41/O/N/21/Q6 [6 Marks]

HARD · Syllabus 30.2

When benzene is reacted with 1-chloropropane in the presence of AlCl_3 , the major product isolated is (1-methylethyl)benzene (cumene) rather than propylbenzene.

- (a) Draw the structural formula of propylbenzene and cumene. [2]

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- (b) Explain why cumene is the major product by referring to carbocation stability and a 1,2-hydride shift. [3]

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- (c) Suggest how propylbenzene could be prepared without rearrangement using Friedel-Crafts acylation. [1]

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13 Chlorination of Arenes: Chlorobenzene vs (Chloromethyl)benzene Tests — 9701/42/M/J/20/Q6 [6 Marks]

HARD · Syllabus 30.3

Two isomeric halogenated arenes have the molecular formula $\text{C}_7\text{H}_7\text{Cl}$:

- Isomer X: 1-chloro-4-methylbenzene (4-chlorotoluene)
- Isomer Y: (chloromethyl)benzene (benzyl chloride)

- (a) Describe a chemical test to distinguish between Isomer X and Isomer Y, stating the reagents, conditions, and observations. [3]

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- (b) Write the ionic equation for the precipitation reaction in the positive test. [1]

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- (c) Explain the difference in reactivity towards hydrolysis between Isomer X and Isomer Y. [2]

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14 Multi-Substituted Arenes: Synthesis of 4-Nitrobenzoic Acid — 9701/41/M/J/20/Q6 [6 Marks]

HARD · Syllabus 30.3

A chemist wishes to synthesize 4-nitrobenzoic acid from methylbenzene.

Two synthetic routes are considered:

- Route 1: Nitration first, followed by oxidation.
- Route 2: Oxidation first, followed by nitration.

- (a) Predict the organic product obtained in Route 2, and explain why Route 2 fails to produce 4-nitrobenzoic acid. [3]

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- (b) Explain why Route 1 successfully yields 4-nitrobenzoic acid, naming the isomers formed in the nitration step. [2]
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- (c) State how 4-nitromethylbenzene can be separated from its 2-nitro isomer. [1]
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15 Halogen Carriers in Electrophilic Substitution: Role of Lewis Acids — 9701/42/O/N/19/Q6 [6 Marks]

HARD · Syllabus 30.2

The chlorination of benzene requires an anhydrous iron(III) chloride catalyst, FeCl_3 .

- (a) Explain what is meant by a Lewis acid, and explain why FeCl_3 acts as a Lewis acid. [2]
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- (b) Write an equation showing how FeCl_3 polarises and cleaves Cl_2 to generate the electrophile. [2]
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- (c) Draw the two-step mechanism for the reaction of benzene with the generated electrophile, showing regeneration of FeCl_3 . [2]
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16 Kekulé Benzene vs 1,3,5-Hexatriene Comparison — 9701/41/O/N/19/Q6 [6 Marks]

HARD · Syllabus 30.1

Benzene (C_6H_6) and hexa-1,3,5-triene ($\text{CH}_2=\text{CH}=\text{CH}=\text{CH}=\text{CH}_2$) both contain three double bonds in their formal formulas.

- (a) Describe what is observed when bromine water is shaken with hexa-1,3,5-triene and with benzene at room temperature. [2]
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- (b) Explain the difference in reactivity towards bromine water between these two compounds. [2]
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- (c) State the number of σ -bonds and π -bonds present in one molecule of benzene. [2]
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SECTION B: 4-MARK STRUCTURED EXAM QUESTIONS (40% TARIFF · Q17–Q32)

17 Bonding in Benzene: sp² Hybridisation & Bond Angles — 9701/42/M/J/23/Q7 [4 Marks]

MEDIUM · Syllabus 30.1

Benzene is a planar, symmetrical hydrocarbon.

- (a) State the hybridisation of the carbon atoms in benzene and the C–C–C bond angle. [2]

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- (b) Describe the shape and origin of the π -electron cloud in benzene. [2]

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18 Equation and Reagents for Nitration of Benzene — 9701/41/M/J/23/Q7 [4 Marks]

MEDIUM · Syllabus 30.2

Nitrobenzene is an important industrial precursor for dyes.

- (a) Write the balanced chemical equation for the nitration of benzene. [2]

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- (b) State the exact reagents and temperature required for mono-nitration. [2]

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19 Friedel–Crafts Methylation of Benzene — 9701/42/O/N/23/Q7 [4 Marks]

MEDIUM · Syllabus 30.2

Methylbenzene is synthesized from benzene.

- (a) State the reagents and catalyst required for this transformation. [2]

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- (b) Name the inorganic gas evolved as a by-product, and write the overall equation. [2]

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20 2,4-Directing Effect of Methyl Group — 9701/41/O/N/23/Q7 [4 Marks]

MEDIUM · Syllabus 30.3

When methylbenzene is brominated in the presence of FeBr₃, two isomers are formed.

- (a) Draw the structural formulas of the two major organic isomers. [2]

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- (b) Explain why substitution occurs predominantly at the 2- and 4-positions rather than the 3-position. [2]

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21 Oxidation of Ethylbenzene with Alkaline KMnO_4 — 9701/42/M/J/22/Q7 [4 Marks]

MEDIUM · Syllabus 30.4

Ethylbenzene, $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_3$, is heated under reflux with alkaline KMnO_4 and then acidified.

- (a) Identify the aromatic organic product formed. [2]

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- (b) State what happens to the remaining carbon of the ethyl chain, naming the product. [2]

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22 Why Benzene Undergoes Substitution Rather than Addition — 9701/41/M/J/22/Q7 [4 Marks]

MEDIUM · Syllabus 30.1

Unlike alkenes, benzene resists addition reactions.

- (a) Explain why benzene does not undergo electrophilic addition with bromine under standard conditions. [2]

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- (b) Explain why electrophilic substitution is thermodynamically preferred. [2]

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23 Halogen Carrier Regeneration in Chlorination of Benzene — 9701/42/O/N/22/Q7 [4 Marks]

MEDIUM · Syllabus 30.2

Aluminium chloride acts as a catalyst in electrophilic substitution.

- (a) Write the equation for the formation of the arenium intermediate when benzene reacts with Cl^+ . [2]

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- (b) Write the equation showing how AlCl_3 is regenerated from AlCl_4^- . [2]

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24 3-Directing Effect of the Nitro Group — 9701/41/O/N/22/Q7 [4 Marks]

MEDIUM · Syllabus 30.3

Nitrobenzene is further nitrated to 1,3-dinitrobenzene.

- (a) Draw the structural formula of 1,3-dinitrobenzene. [2]

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- (b) Explain why the nitro group is described as a deactivating group. [2]

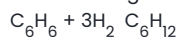
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25 Hydrogenation of Benzene to Cyclohexane — 9701/42/M/J/21/Q7 [4 Marks]

MEDIUM · Syllabus 30.1

Under forcing conditions, benzene can be hydrogenated to cyclohexane:



(a) State the catalyst and the temperature and pressure conditions required. [2]

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(b) State the change in hybridisation of the carbon atoms during this reaction. [2]

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26 Comparison of Reactivity: Benzene vs Methylbenzene — 9701/41/M/J/21/Q7 [4 Marks]

MEDIUM · Syllabus 30.3

Methylbenzene is nitrated approximately 25 times faster than benzene.

(a) Explain why methylbenzene reacts faster than benzene in electrophilic substitution. [2]

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(b) State the temperature required to mono-nitrate methylbenzene compared to benzene. [2]

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27 Side-Chain Chlorination of Methylbenzene by Free Radicals — 9701/42/O/N/21/Q7 [4 Marks]

MEDIUM · Syllabus 30.3

Boiling methylbenzene with chlorine in UV light yields (chloromethyl)benzene.

(a) Name the type of bond fission that initiates this reaction. [2]

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(b) State the products formed if chlorine gas is bubbled in excess. [2]

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28 Synthesis of Phenylethene (Styrene) — 9701/41/O/N/21/Q7 [4 Marks]

MEDIUM · Syllabus 30.4

Phenylethene is the monomer for polystyrene.

(a) Show how phenylethene, $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$, can be prepared from ethylbenzene. [2]

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(b) Describe a chemical test to confirm the presence of the alkene double bond in phenylethene. [2]

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29 Friedel-Crafts Acylation of Methylbenzene — 9701/42/M/J/20/Q7 [4 Marks]

MEDIUM · Syllabus 30.2

Methylbenzene reacts with ethanoyl chloride in the presence of AlCl_3 .

- (a) Identify the major organic product formed, giving its IUPAC name. [2]

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- (b) Explain why 4-methylphenylethanone is formed in preference to 2-methylphenylethanone. [2]

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30 Combustion Hazard of Benzene in Industry — 9701/41/M/J/20/Q7 [4 Marks]

MEDIUM · Syllabus 30.1

Benzene is a volatile, carcinogenic liquid.

- (a) State two health or safety hazards associated with handling benzene in the laboratory. [2]

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- (b) Name a safer aromatic hydrocarbon frequently used as a laboratory substitute for benzene. [2]

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31 Identification of Chiral Center in Aromatic Derivatives — 9701/42/O/N/19/Q7 [4 Marks]

MEDIUM · Syllabus 30.3

Consider the aromatic compounds: (chloromethyl)benzene, 1-phenylethanol, 2-phenylethanol.

- (a) Identify which of these compounds exhibits optical activity. [2]

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- (b) Draw the displayed formula of this compound and circle its chiral carbon. [2]

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32 Resistance of Benzene Ring to Oxidation — 9701/41/O/N/19/Q7 [4 Marks]

MEDIUM · Syllabus 30.4

When methylbenzene is boiled with acidified potassium dichromate, the methyl group is oxidised but the ring remains intact.

- (a) Explain why the benzene ring is exceptionally resistant to chemical oxidising agents. [2]

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- (b) State what reagent can rupture the benzene ring completely. [2]

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SECTION C: 2-MARK TARGETED EXAM QUESTIONS (20% TARIFF · Q33–Q40)

33 Carbon–Carbon Bond Length in Benzene — 9701/42/M/J/23/Q9 [2 Marks]

EASY · Syllabus 30.1

X-ray diffraction determines bond lengths accurately.

- (a) State the experimental C–C bond length in benzene and compare it with standard C–C single and C=C double bonds. [2]

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34 Electrophile in Nitration of Benzene — 9701/41/M/J/23/Q9 [2 Marks]

EASY · Syllabus 30.2

Electrophilic substitution involves reactive electrophiles.

- (a) Name and give the chemical formula of the active electrophile in the nitration of benzene. [2]

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35 Role of AlCl_3 in Halogenation — 9701/42/O/N/23/Q9 [2 Marks]

EASY · Syllabus 30.2

Halogen carriers facilitate electrophilic substitution.

- (a) State the role of anhydrous AlCl_3 in the chlorination of benzene. [2]

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36 Directing Effect of Alkyl Groups — 9701/41/O/N/23/Q9 [2 Marks]

EASY · Syllabus 30.3

Substituents direct electrophilic substitution.

- (a) State the directing effect of an alkyl group on a benzene ring. [2]

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37 Oxidation Product of Methylbenzene — 9701/42/M/J/22/Q9 [2 Marks]

EASY · Syllabus 30.4

Alkyl side-chains undergo vigorous oxidation.

- (a) Name and draw the formula of the organic compound formed when methylbenzene is refluxed with alkaline KMnO_4 and then acidified. [2]

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38 Resonance Energy of Benzene Value — 9701/41/M/J/22/Q9 [2 Marks]

EASY · Syllabus 30.1

Thermochemical data quantify aromatic stability.

- (a) State the numerical value of the resonance stabilization energy of benzene, with units. [2]

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39 Conditions for Side-Chain Chlorination of Toluene — 9701/42/O/N/22/Q9 [2 Marks]

EASY · Syllabus 30.3

Different conditions change substitution site.

- (a) State the conditions required to chlorinate the methyl group of methylbenzene rather than the aromatic ring.

[2]

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40 Wheland Intermediate Definition — 9701/41/O/N/22/Q9 [2 Marks]

EASY · Syllabus 30.2

Arenium ions are key intermediates.

- (a) Describe the structure of the carbocation intermediate formed during electrophilic substitution of benzene.

[2]

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SECTION D: HIGH-FREQUENCY CORE REPEATS — 10 MOST FREQUENTLY TESTED QUESTIONS (Q41–Q50)

41 [CORE REPEAT 1] Complete Enthalpy of Hydrogenation & Delocalisation Analysis — 9701/42/M/J/23/Q6 [6 Marks] · Syllabus 30.1

The experimental enthalpy of hydrogenation of benzene is compared to the theoretical value in Fig. 41.1.

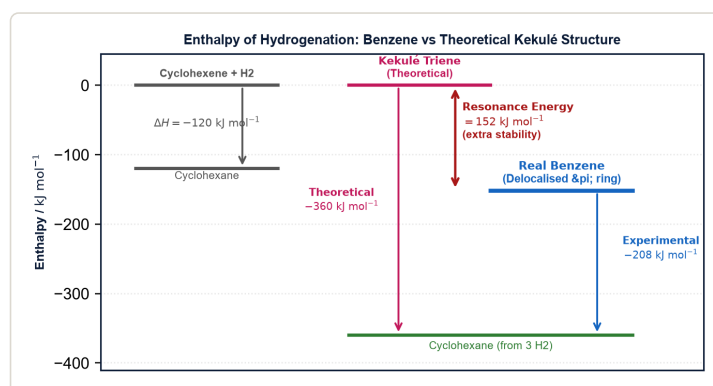


Fig. 41.1: Thermochemical comparison of benzene and theoretical cyclohexa-1,3,5-triene.

- (a) Calculate the resonance stabilization energy of benzene using $H^{\circ}_{\text{hyd}}(\text{cyclohexene}) = -120 \text{ kJ mol}^{-1}$ and $H^{\circ}_{\text{hyd}}(\text{benzene}) = -208 \text{ kJ mol}^{-1}$.

[2]

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- (b) Explain how the delocalised π -electron cloud accounts for the extra stability of benzene.

[2]

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- (c) State two pieces of structural evidence that support the delocalised model over the Kekulé model.

[2]

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42 [CORE REPEAT 2] Full Mechanism of Nitration of Benzene — 9701/41/M/J/23/Q6 [6 Marks]

HARD · Syllabus 30.2

Nitration of benzene is a classic electrophilic aromatic substitution, illustrated in Fig. 42.1.

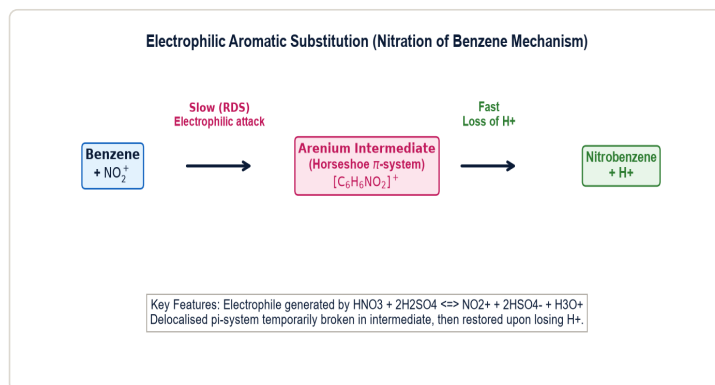


Fig. 42.1: Stepwise electrophilic substitution pathway for benzene nitration.

- (a) Write the balanced chemical equation for the generation of NO_2^+ from concentrated HNO_3 and concentrated H_2SO_4 . [2]
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- (b) Draw the mechanism using curly arrows showing the arenium carbocation intermediate and loss of H^+ . [3]
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- (c) State why sulfuric acid is regenerated at the end of the reaction. [1]
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43 [CORE REPEAT 3] Ring vs Side-Chain Chlorination of Methylbenzene — 9701/42/O/N/23/Q6 [6 Marks]

HARD · Syllabus 30.3

The site of substitution in methylbenzene is controlled by the reaction conditions.

- (a) State the reagents, catalyst, and conditions to chlorinate the ring, naming the two organic products. [3]
-
-
-
- (b) State the conditions to chlorinate the methyl side-chain, naming the product. [2]
-
-
- (c) State the mechanism type for the ring reaction and for the side-chain reaction. [1]
-
-

44 [CORE REPEAT 4] Side-Chain Oxidation of Alkylbenzenes & Decolourisation Test — 9701/41/O/N/23/Q6 [6 Marks]

HARD · Syllabus 30.4

Alkyl side-chains attached to benzene rings are oxidised by acidified KMnO_4 .

- (a) Write the balanced chemical equation for the oxidation of methylbenzene to benzoic acid using $[O]$ as the oxidising agent. [2]
-
-
- (b) State the colour change observed in the reaction flask when hot $KMnO_4$ is reduced. [2]
-
-
- (c) Explain why 1,4-dimethylbenzene yields a dicarboxylic acid, and name the commercial polymer made from this dicarboxylic acid. [2]
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45 [CORE REPEAT 5] Friedel-Crafts Acylation of Benzene with Ethanoyl Chloride — 9701/42/M/J/22/Q6 [4 Marks] MEDIUM · Syllabus 30.2

Ethanoyl chloride reacts with benzene in the presence of $AlCl_3$.

- (a) Name the organic product formed and state the functional group present. [2]
-
-
- (b) Write the equation for the formation of the electrophile. [2]
-
-

46 [CORE REPEAT 6] Directing Effects of -OH vs -NO₂ Groups — 9701/41/M/J/22/Q6 [4 Marks] MEDIUM · Syllabus 30.3

Substituents alter the orientation of further electrophilic attack.

- (a) State the directing effect and activating/deactivating nature of the -OH group in phenol. [2]
-
-
- (b) State the directing effect and activating/deactivating nature of the -NO₂ group in nitrobenzene. [2]
-
-

47 [CORE REPEAT 7] Inertness of Chlorobenzene to Hydrolysis — 9701/42/O/N/22/Q6 [4 Marks] MEDIUM · Syllabus 30.2

Chlorobenzene does not react with aqueous NaOH under reflux.

- (a) Explain why the C-Cl bond in chlorobenzene is stronger than in chloroethane. [2]
-
-
- (b) State why the electron-rich benzene ring repels incoming hydroxide nucleophiles. [2]
-
-

48 [CORE REPEAT 8] Distinguishing Benzene from Alkenes by Bromine Water — 9701/41/O/N/22/Q6 [4 Marks] MEDIUM · Syllabus 30.1

Cyclohexene and benzene are both six-membered carbon rings.

- (a) Describe a simple chemical test using bromine to distinguish between cyclohexene and benzene. [2]

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- (b) Explain why cyclohexene undergoes addition readily while benzene does not. [2]

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49 [CORE REPEAT 9] Planar Hexagonal Symmetry of Benzene — 9701/42/M/J/21/Q6 [2 Marks]

EASY · Syllabus 30.1

Benzene has a regular hexagonal structure.

- (a) State the C–C–C bond angle and geometry of the benzene molecule. [2]

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50 [CORE REPEAT 10] Benzylic Hydrogen Requirement for Side-Chain Oxidation — 9701/41/M/J/21/Q6 [2 Marks] EASY · Syllabus 30.4

Side-chain oxidation requires specific structural features.

- (a) State the structural feature an alkyl side-chain must possess to be oxidised by hot alkaline KMnO_4 to benzoic acid. [2]

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Mark Scheme — Topic 30 — Hydrocarbons (Arenes)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — PAPER 4 MARKING CRITERIA

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
1(a)	Theoretical $H = 3 \times (-120) = -360 \text{ kJ mol}^{-1}$ [1]; Resonance energy = $-360 - (-208) = -152 \text{ kJ mol}^{-1}$ (benzene is 152 kJ mol^{-1} more stable than predicted) [1].	2
1(b)	Each carbon atom is sp^2 hybridised, forming three planar σ -bonds (two C-C, one C-H) at 120° bond angles [1]; The remaining unhybridised 2p atomic orbitals on all six carbon atoms overlap sideways above and below the ring plane to form a continuous delocalised π -electron cloud (donut rings) [1].	2
1(c)	1. All six C-C bond lengths are completely identical (0.139 nm), intermediate between C-C single (0.154 nm) and C=C double (0.134 nm) [1]; 2. Benzene undergoes electrophilic substitution rather than electrophilic addition and does not decolourise bromine water at room temperature [1].	2
2(a)	$\text{HNO}_3 + 2\text{H}_2\text{SO}_4 \rightarrow \text{NO}_2^+ + 2\text{HSO}_4^- + \text{H}_3\text{O}^+$ (or $\text{HNO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{NO}_2^+ + \text{HSO}_4^- + \text{H}_2\text{O}$) [1]; H_2SO_4 acts as a Brønsted-Lowry acid / catalyst by protonating HNO_3 [1].	2
2(b)	Curly arrow from delocalised π -ring to NO_2^+ [1]; Correct horseshoe arenium intermediate $[\text{C}_6\text{H}_6\text{NO}_2]^+$ with opening towards substituted carbon and positive charge inside [1]; Curly arrow from C-H bond into ring to restore π -system, and H^+ accepted by HSO_4^- to reform H_2SO_4 [1].	3
2(c)	To prevent multiple nitrations (formation of 1,3-dinitrobenzene and 1,3,5-trinitrobenzene) [1].	1
3(a)	$\text{CH}_3\text{COCl} + \text{AlCl}_3 \rightarrow \text{CH}_3\text{CO}^+ + \text{AlCl}_4^-$ [2].	2
3(b)	Curly arrow from benzene ring to CH_3CO^+ ; horseshoe arenium intermediate formed; curly arrow from C-H bond back into ring; AlCl_4^- takes H^+ to give $\text{HCl} + \text{AlCl}_3$ [2].	2
3(c)	The acyl group ($-\text{COCH}_3$) is strongly electron-withdrawing ($-I$ and $-M$), deactivating the ring towards further electrophilic attack [1]; Alkyl groups ($-R$) are electron-donating ($+I$), making the mono-alkylated ring more electron-rich and more reactive than benzene, encouraging multiple alkylations [1].	2
4(a)	2-chloromethylbenzene (o-chlorotoluene) and 4-chloromethylbenzene (p-chlorotoluene) [1]; Electrophilic aromatic substitution [1]; The methyl group is electron-donating ($+I$ effect) and activates the 2- and 4-positions by stabilising the carbocation intermediate at those positions [1].	3
4(b)	(Chloromethyl)benzene, $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ [1]; Free-radical substitution of the alkyl side-chain [1].	2
4(c)	Step 1: $\text{C}_6\text{H}_5\text{CH}_3 + \text{Cl}^\bullet \rightarrow \text{C}_6\text{H}_5\text{CH}_2^\bullet + \text{HCl}$; Step 2: $\text{C}_6\text{H}_5\text{CH}_2^\bullet + \text{Cl}_2 \rightarrow \text{C}_6\text{H}_5\text{CH}_2\text{Cl} + \text{Cl}^\bullet$ [1].	1
5(a)	Purple solution decolourises and a brown/black precipitate of MnO_2 forms [1]; Upon acidification, a white crystalline precipitate of benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$) forms [1].	2
5(b)	Benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$ [1]; Carbon dioxide, CO_2 (or carbonate) [1].	2
5(c)	Side-chain oxidation requires at least one benzylic hydrogen atom attached to the carbon directly bonded to the benzene ring [1]; In tert-butylbenzene, the benzylic carbon is tertiary and bonded to three methyl groups (zero benzylic C-H bonds), making it resistant to oxidative attack [1].	2
6(a)	Benzene < Methylbenzene < Phenol [1].	1
6(b)	Br_2 with anhydrous FeBr_3 (or AlBr_3) catalyst at warm temperature [1]; $\text{C}_6\text{H}_6 + \text{Br}_2 \rightarrow \text{C}_6\text{H}_5\text{Br} + \text{HBr}$ [1].	2
6(c)	In phenol, the lone pair on the oxygen atom of the $-\text{OH}$ group delocalises into the benzene π -ring [1]; This significantly increases the π -electron density of the ring, activating it [1]; The electron-rich ring is capable of polarising non-polar Br_2 molecules without needing a halogen carrier catalyst, whereas benzene's stable delocalised ring requires FeBr_3 to generate the powerful Br^+ electrophile [1].	3
7(a)	Nitrogen has a formal positive charge and is bonded to two electronegative oxygens, exerting a strong electron-withdrawing inductive and resonance ($-I$ and $-M$) effect [1]; This withdraws electron density from the π -system, dramatically deactivating the entire ring [1]; Electron density is depleted most severely at the 2- and 4-positions, making attack at the 3-position the pathway of lowest relative activation energy [1].	3
7(b)	1,3-dinitrobenzene drawn correctly [1].	1
7(c)	Benzene requires conc. $\text{HNO}_3 + \text{conc. H}_2\text{SO}_4$ at $50-55^\circ\text{C}$ [1]; Nitrobenzene is deactivated and requires much harsher conditions: reflux at 100°C (or fuming nitric acid) [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
8(a)	1-bromo-2-ethylbenzene (2-bromoethylbenzene) and 1-bromo-4-ethylbenzene (4-bromoethylbenzene) [2].	2
8(b)	C ₆ H ₅ -C*(Br)-CH ₃ drawn with asterisk on C1 [1]; Carbon-1 is bonded to four different groups (-H, -Br, -CH ₃ , -C ₆ H ₅) [1].	2
8(c)	Abstraction of H from C1 forms a benzylic free radical, C ₆ H ₅ -C•H-CH ₃ [1]; The unpaired electron is delocalised into the aromatic π-system, conferring extraordinary stability compared to the primary alkyl radical at C2 [1].	2
9(a)	C ₆ H ₆ (l) + 7.5O ₂ (g) → 6CO ₂ (g) + 3H ₂ O(l) (or 2C ₆ H ₆ + 15O ₂ → 12CO ₂ + 6H ₂ O) [2].	2
9(b)	Benzene: Mr = 78.0 &Arr; %C = (72.0 / 78.0) × 100 = 92.3% [1]; Hexane: Mr = 86.0 &Arr; %C = (72.0 / 86.0) × 100 = 83.7% [1].	2
9(c)	Benzene has an exceptionally high carbon-to-hydrogen ratio (1:1) [1]; There is insufficient oxygen during open-air burning to completely oxidise all carbon, resulting in incomplete combustion that liberates glowing solid incandescent carbon particles (soot) [1].	2
10(a)	Chloromethane (CH ₃ Cl) with anhydrous aluminium chloride (AlCl ₃) catalyst at room temperature (Friedel-Crafts alkylation) [2].	2
10(b)	Reflux with alkaline potassium manganate(VII) (KMnO ₄ + NaOH), followed by acidification with dilute sulfuric acid (or dilute HCl) [2].	2
10(c)	Involves more synthetic steps, requires moisture-sensitive anhydrous ether for Grignard reagents, and halogenobenzene has an unreactive C-Br bond due to π-delocalisation [2].	2
11(a)	CH ₃ CH ₂ Cl + OH ⁻ → CH ₃ CH ₂ OH + Cl ⁻ [1].	1
11(b)	A lone pair of electrons on the chlorine atom overlaps with the delocalised π-electron system of the benzene ring [1]; This imparts partial double bond character to the C-Cl bond [1]; The bond is shorter, stronger, and has a much higher bond dissociation enthalpy, resisting cleavage [1].	3
11(c)	The incoming nucleophile (OH ⁻) is negatively charged and is electrostatically repelled by the high electron density of the delocalised π-electron cloud above and below the benzene ring [2].	2
12(a)	Propylbenzene: C ₆ H ₅ -CH ₂ CH ₂ CH ₃ ; Cumene: C ₆ H ₅ -CH(CH ₃) ₂ [2].	2
12(b)	Reaction generates an initial primary carbocation: CH ₃ CH ₂ C ⁺ H ₂ [1]; A 1,2-hydride shift occurs: a hydrogen atom with its pair of electrons migrates from C ₂ to C ₁ , transforming the primary carbocation into a more stable secondary carbocation: CH ₃ C ⁺ H-CH ₃ [1]; Secondary carbocation is stabilised by the positive inductive (+I) effect of two methyl groups, attacking benzene much faster [1].	3
12(c)	React benzene with propanoyl chloride (CH ₃ CH ₂ COCl) and AlCl ₃ to give C ₆ H ₅ COCH ₂ CH ₃ , then reduce the carbonyl group using Clemmensen reduction (Zn/HCl) or Wolff-Kishner reduction [1].	1
13(a)	Warm each isomer with aqueous sodium hydroxide (or ethanol/water), cool, acidify with dilute nitric acid (HNO ₃), and add aqueous silver nitrate (AgNO ₃) [2]; Isomer Y ((chloromethyl)benzene) gives an immediate white precipitate of AgCl; Isomer X (4-chlorotoluene) produces no precipitate [1].	3
13(b)	Ag ⁺ (aq) + Cl ⁻ (aq) → AgCl(s) [1].	1
13(c)	In Isomer Y, the chlorine is on an aliphatic side-chain (benzylic) and hydrolyses easily; in Isomer X, chlorine is attached directly to the aromatic ring where lone pair delocalisation strengthens the C-Cl bond [2].	2
14(a)	Route 2 produces 3-nitrobenzoic acid [1]; The carboxylic acid group (-COOH) is electron-withdrawing and is a 3-directing (meta-directing) group [1]; Nitration of benzoic acid directs the incoming NO ₂ ⁺ to the 3-position, not the 4-position [1].	3
14(b)	In Route 1, the methyl group (-CH ₃) in methylbenzene is electron-donating (+I) and is a 2,4-directing group [1]; Nitration gives a mixture of 2-nitromethylbenzene and 4-nitromethylbenzene; subsequent oxidation of 4-nitromethylbenzene yields 4-nitrobenzoic acid [1].	2
14(c)	Fractional distillation (or fractional crystallisation) [1].	1
15(a)	A lone-pair electron acceptor [1]; Fe in FeCl ₃ has vacant d-orbitals and an incomplete octet capable of accepting a lone pair from a chlorine molecule [1].	2
15(b)	Cl ₂ + FeCl ₃ → Cl ⁺ + FeCl ₄ ⁻ (or Cl-Cl---FeCl ₃ complex with δ+ on outer Cl) [2].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
15(c)	Electrophilic attack by Cl^+ on benzene forming $[\text{C}_6\text{H}_6\text{Cl}]^+$ arenium intermediate; loss of H^+ to FeCl_4^- yielding $\text{C}_6\text{H}_5\text{Cl} + \text{HCl} + \text{FeCl}_3$ [2].	2
16(a)	Hexa-1,3,5-triene: orange bromine water decolourises immediately / turns colourless [1]; Benzene: orange bromine water remains unchanged (no reaction) [1].	2
16(b)	Hexa-1,3,5-triene contains localised $\text{C}=\text{C}$ double bonds with high localized π -electron density that easily polarise Br_2 and undergo electrophilic addition [1]; Benzene possesses a continuous delocalised π -electron aromatic ring with high resonance stabilisation energy (152 kJ mol^{-1}) that would be permanently destroyed by addition [1].	2
16(c)	12 σ -bonds (six $\text{C}-\text{C}$ σ -bonds and six $\text{C}-\text{H}$ σ -bonds) [1]; 3 π -bonds (6 delocalised π -electrons forming 3 pairs) [1].	2
17(a)	sp^2 hybridisation [1]; 120° [1].	2
17(b)	Formed by sideways overlap of unhybridised 2p atomic orbitals on all six carbon atoms [1]; Two continuous ring-shaped clouds (toroids) of delocalised π -electrons lying directly above and below the planar carbon ring [1].	2
18(a)	$\text{C}_6\text{H}_6 + \text{HNO}_3 \rightarrow \text{C}_6\text{H}_5\text{NO}_2 + \text{H}_2\text{O}$ [2].	2
18(b)	Concentrated nitric acid (HNO_3) and concentrated sulfuric acid (H_2SO_4) [1]; $50-55^\circ\text{C}$ [1].	2
19(a)	Chloromethane, CH_3Cl [1]; Anhydrous aluminium chloride catalyst, AlCl_3 [1].	2
19(b)	Hydrogen chloride gas, HCl [1]; $\text{C}_6\text{H}_6 + \text{CH}_3\text{Cl} \rightarrow \text{C}_6\text{H}_5\text{CH}_3 + \text{HCl}$ [1].	2
20(a)	2-bromomethylbenzene and 4-bromomethylbenzene drawn correctly [2].	2
20(b)	The methyl group is electron-donating by positive inductive (+I) effect [1]; Electrophilic attack at 2- and 4-positions produces carbocation intermediates that have greater resonance stability (tertiary carbocation contributor) [1].	2
21(a)	Benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$ [2].	2
21(b)	It is completely oxidised to carbon dioxide (CO_2) / carbonate [2].	2
22(a)	Electrophilic addition would permanently break the continuous delocalised π -system, losing the 152 kJ mol^{-1} resonance stabilisation energy [2].	2
22(b)	Electrophilic substitution proceeds with the loss of H^+ in the second step, completely regenerating and restoring the stable delocalised aromatic π -system [2].	2
23(a)	$\text{C}_6\text{H}_6 + \text{Cl}^+ \rightarrow [\text{C}_6\text{H}_6\text{Cl}]^+$ [2].	2
23(b)	$[\text{C}_6\text{H}_6\text{Cl}]^+ + \text{AlCl}_4^- \rightarrow \text{C}_6\text{H}_5\text{Cl} + \text{HCl} + \text{AlCl}_3$ [2].	2
24(a)	Benzene ring with $-\text{NO}_2$ groups at positions 1 and 3 drawn correctly [2].	2
24(b)	The strongly electronegative nitro group pulls electron density away from the ring via inductive and resonance effects [1]; Lowering ring electron density and making it a poorer nucleophile towards electrophiles [1].	2
25(a)	Finely divided nickel catalyst (or platinum/palladium) [1]; $150-200^\circ\text{C}$ and high pressure (approx. 30 atm) [1].	2
25(b)	sp^2 hybridisation in planar benzene changes to sp^3 hybridisation in non-planar cyclohexane [2].	2
26(a)	The methyl group pushes electron density into the ring through a positive inductive (+I) effect [1]; Increasing the π -electron density and making the ring more susceptible to electrophilic attack [1].	2
26(b)	Around 30°C (milder temperature than benzene's $50-55^\circ\text{C}$) [2].	2
27(a)	Homolytic fission of the $\text{Cl}-\text{Cl}$ bond by UV light [2].	2
27(b)	(Dichloromethyl)benzene, $\text{C}_6\text{H}_5\text{CHCl}_2$, and (trichloromethyl)benzene, $\text{C}_6\text{H}_5\text{CCl}_3$ [2].	2
28(a)	Catalytic dehydrogenation over iron(III) oxide catalyst at 600°C (or brominate with Br_2/UV to $\text{C}_6\text{H}_5\text{CH}(\text{Br})\text{CH}_3$ then eliminate with ethanolic KOH) [2].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
28(b)	Add bromine water: orange bromine water decolourises immediately [2].	2
29(a)	4-methylphenylethanone (1-(4-methylphenyl)ethan-1-one) [2].	2
29(b)	The bulky -COCH_3 acyl group and the -CH_3 methyl group experience steric hindrance/crowding at the adjacent 2-position, making attack at the open 4-position sterically favoured [2].	2
30(a)	1. Known human carcinogen / causes leukaemia [1]; 2. Highly flammable volatile liquid (fire/explosion risk) [1].	2
30(b)	Methylbenzene (toluene) [2] (its methyl group allows rapid metabolic excretion via benzoic acid, making it far less toxic).	2
31(a)	1-phenylethanol, $\text{C}_6\text{H}_5\text{CH}(\text{OH})\text{CH}_3$ [2].	2
31(b)	Displayed formula with circle around C1 bonded to -H , -OH , -CH_3 , and $\text{-C}_6\text{H}_5$ [2].	2
32(a)	The delocalised π -electron system possesses a high resonance stabilisation energy of 152 kJ mol^{-1} , which makes breaking the aromatic carbon skeleton energetically prohibitive [2].	2
32(b)	Ozone (ozonolysis) or combustion in excess oxygen / catalytic oxidation at 450°C over V_2O_5 to maleic anhydride [2].	2
33(a)	0.139 nm [1]; Intermediate between single bond (0.154 nm) and double bond (0.134 nm) [1].	2
34(a)	Nitronium ion [1]; NO_2^+ [1].	2
35(a)	Halogen carrier / Lewis acid catalyst [1]; Polarises the chlorine molecule to generate the Cl^+ electrophile [1].	2
36(a)	2- and 4-directing (ortho- and para-directing) [1]; and activating [1].	2
37(a)	Benzoic acid [1]; $\text{C}_6\text{H}_5\text{COOH}$ [1].	2
38(a)	152 kJ mol^{-1} [2].	2
39(a)	Chlorine gas (Cl_2), boiling / high temperature [1]; and ultraviolet (UV) light (absence of Lewis acid) [1].	2
40(a)	A non-aromatic cyclohexadienyl cation (arenium ion) [1]; with a delocalised 4π -electron system spread over 5 carbon atoms (horseshoe) carrying a +1 charge [1].	2
41(a)	Theoretical $H = 3 \times (-120) = -360 \text{ kJ mol}^{-1}$ [1]; Resonance energy = $-360 - (-208) = -152 \text{ kJ mol}^{-1}$ [1].	2
41(b)	Sideways overlap of 2p orbitals spreads 6 π -electrons over all six carbon nuclei [1]; This delocalisation lowers electron-electron repulsion and maximizes electron-nuclear attraction, creating a low-energy, highly stable aromatic ring [1].	2
41(c)	1. All six C-C bonds have equal length (0.139 nm) [1]; 2. Benzene is a regular planar hexagon with 120° bond angles (no alternating short and long bonds) [1].	2
42(a)	$\text{HNO}_3 + 2\text{H}_2\text{SO}_4 \rightarrow \text{NO}_2^+ + 2\text{HSO}_4^- + \text{H}_3\text{O}^+$ [2].	2
42(b)	Curly arrow from π -ring to NO_2^+ [1]; Horseshoe carbocation intermediate drawn with + charge and H, NO_2 attached [1]; Curly arrow from C-H bond back into ring to regenerate delocalised π -system [1].	3
42(c)	H^+ released from intermediate combines with HSO_4^- to reform H_2SO_4 , confirming its role as a catalyst [1].	1
43(a)	Cl_2 with anhydrous AlCl_3 (or FeCl_3) in the dark at room temperature [1]; 2-chloromethylbenzene and 4-chloromethylbenzene [2].	3
43(b)	Cl_2 gas, boiling / heat, under ultraviolet (UV) light [1]; (Chloromethyl)benzene [1].	2
43(c)	Ring: Electrophilic aromatic substitution; Side-chain: Free-radical substitution [1].	1
44(a)	$\text{C}_6\text{H}_5\text{CH}_3 + 3[\text{O}] \rightarrow \text{C}_6\text{H}_5\text{COOH} + \text{H}_2\text{O}$ [2].	2
44(b)	Purple solution decolourises (or forms a brown precipitate of MnO_2) [2].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
44(c)	Both benzylic methyl groups are oxidised to give benzene-1,4-dicarboxylic acid (terephthalic acid) [1]; Used to manufacture Terylene / PET polyester [1].	2
45(a)	Phenylethanone (acetophenone) [1]; Ketone ($\text{C}=\text{O}$) [1].	2
45(b)	$\text{CH}_3\text{COCl} + \text{AlCl}_3 \rightarrow \text{CH}_3\text{CO}^+ + \text{AlCl}_4^-$ [2].	2
46(a)	2,4-directing (ortho/para) [1]; Activating [1].	2
46(b)	3-directing (meta) [1]; Deactivating [1].	2
47(a)	Overlap of a chlorine p-orbital lone pair with the delocalised π -system of the ring gives the C-Cl bond partial double-bond character [2].	2
47(b)	The delocalised π -electron cloud above and below the ring creates a region of high negative charge density, electrostatically repelling negative OH^- ions [2].	2
48(a)	Add bromine water (or Br_2 in CCl_4) in the dark at room temperature: cyclohexene decolourises bromine water immediately; benzene produces no colour change [2].	2
48(b)	Cyclohexene has a localized $\text{C}=\text{C}$ double bond with high π -electron density that polarises Br_2 ; benzene's electrons are delocalised and stabilized by resonance, resisting addition to preserve aromaticity [2].	2
49(a)	Bond angle: 120° [1]; Geometry: planar regular hexagon [1].	2
50(a)	At least one hydrogen atom bonded directly to the benzylic carbon (the carbon atom attached to the benzene ring) [2].	2